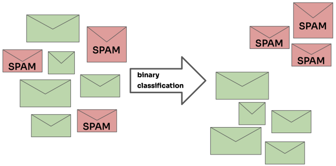
Introduction to Machine Learning

Classification
Tasks

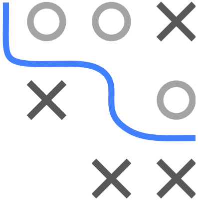


Learning goals

- Classification is supervised learning with categorical labels
- Binary vs. multiclass
- Some examples of classification tasks

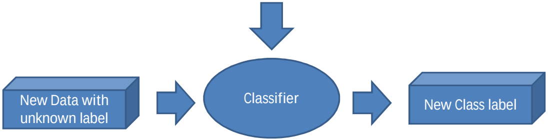
CLASSIFICATION

Learn functions that assign class labels to observation / feature vectors. Each observation belongs to exactly one class. The main difference to regression is that the target is categorical.



Our Data

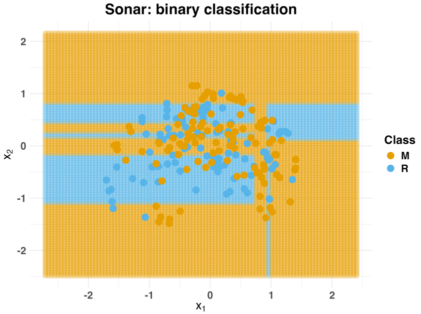
Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.1	3.5	1.4	0.2	setosa
5.9	3.0	5.1	1.8	virginica

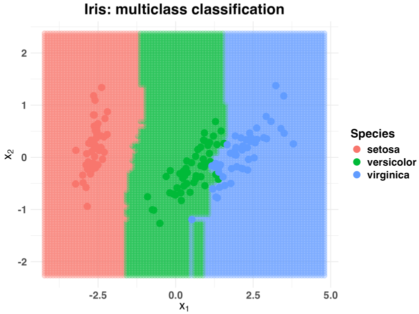


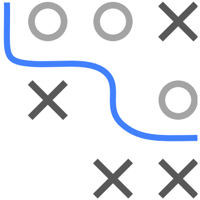
Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.4	3.3	3.2	1.1	???

BINARY AND MULTICLASS TASKS

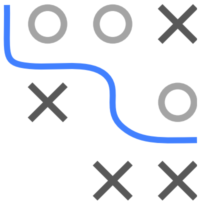
Tasks have a finite number of (unordered) classes.
They can be **binary** or **multiclass**.



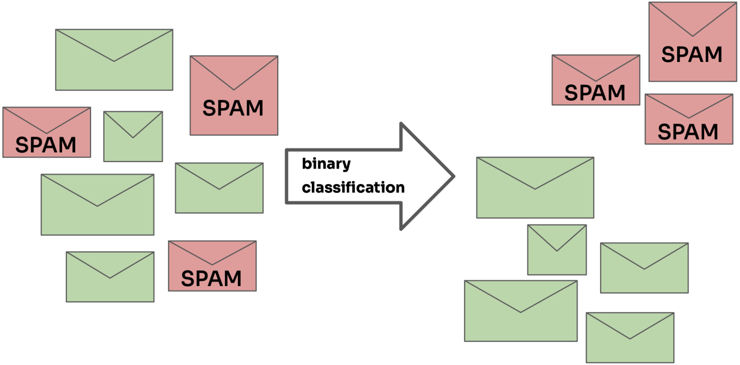




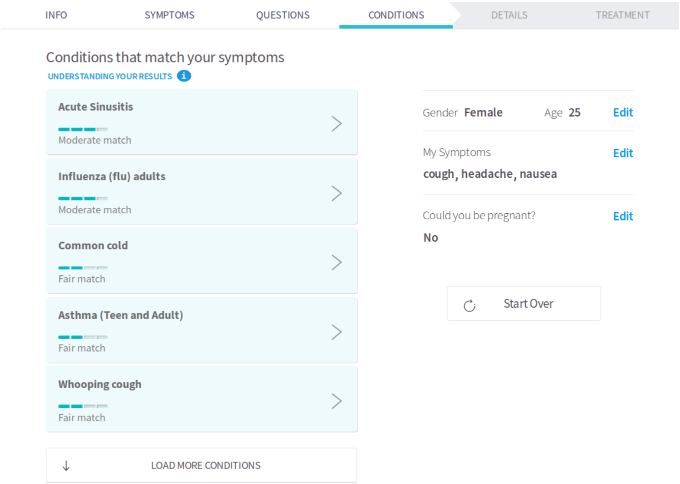
BINARY CLASSIFICATION TASK - EXAMPLES



- Credit risk prediction, based on personal data and transactions
- Spam detection, based on textual features
- Churn prediction, based on customer behavior
- Predisposition for specific illness, based on genetic data



MULTICLASS TASK - MEDICAL DIAGNOSIS



https://symptoms.webmd.com

MULTICLASS TASK - IRIS

The iris dataset was introduced by the statistician Ronald Fisher and is one of the most frequent used data sets. Originally, it was designed for linear discriminant analysis.

Setosa

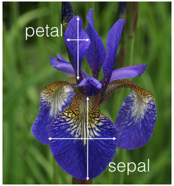
Versicolor

Virginica

Source:
https://en.wikipedia.org/wiki/Iris_flower_data_set

MULTICLASS TASK - IRIS

- 150 iris flowers
- Predict subspecies
- Based on sepal and petal length / width in [cm]




```
##      Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
##   1:          5.1         3.5          1.4         0.2    setosa
##   2:          4.9         3.0          1.4         0.2    setosa
##   3:          4.7         3.2          1.3         0.2    setosa
##   4:          4.6         3.1          1.5         0.2    setosa
##   5:          5.0         3.6          1.4         0.2    setosa
##  ---
## 146:          6.7         3.0          5.2         2.3 virginica
## 147:          6.3         2.5          5.0         1.9 virginica
## 148:          6.5         3.0          5.2         2.0 virginica
## 149:          6.2         3.4          5.4         2.3 virginica
## 150:          5.9         3.0          5.1         1.8 virginica
```

MULTICLASS TASK - IRIS